

Name:

Module 4 Assignment: Forensic Ballistics

/42

Lesson A: Firearm Basics

1. Describe two differences between a bullet and a cartridge. (2 marks)

- A bullet is one of several parts of a cartridge. (1 mark)
- When a gun is fired, the bullet is propelled from the barrel and the empty cartridge casing remains behind. (1 mark)

2. Discuss why bullets and their casings are often found in different locations at crime scenes. (2 marks)

When a firearm is discharged, the bullet is fired from the gun barrel by the explosive gases released by the ignition of gunpowder in the shell casing. Then, the casing is either retained in the firearm or is ejected by the force of the recoil that automatically chambers a new cartridge in the process. Therefore, while the bullet may be lodged in an object at the crime scene, the shell casing may still be in the firearm or may have been ejected, landing somewhere in the immediate area of the shooter's position.

3. Using your knowledge of velocity, kinetic energy, and trajectory, explain how a handgun round becomes less effective than a rifle bullet over long distances. (3 marks)

A rifle bullet will leave the rifle with greater kinetic energy and velocity because a larger amount of gunpowder is contained in the cartridge. A handgun round typically suffers from a significant reduction in velocity and kinetic energy as it travels beyond a certain distance. This means a drop in trajectory. A bullet fired from a large calibre rifle retains a large amount of velocity and kinetic energy even after it has travelled one or two kilometres. This enables the bullet to have a flatter trajectory. In contrast, the kinetic energy of a handgun round is expended much more quickly. Consequently, this round will lose height and fall to the ground much sooner.

4. Explain why a police tactical unit always takes at least one assault rifle or sniper rifle when responding to a call involving a suspect armed with a gun. (2 marks)

A high-powered rifle is able to fire a bullet over long distances with a high degree of accuracy and kinetic energy. This provides officers with an opportunity to effectively neutralize a target with reduced risk to the public as well as other officers. A scope mounted on the gun allows for more accurate aiming.

Name:

Lesson B: Ballistic Fingerprinting and Wound Ballistics

5. Identify three unique characteristics that a forensic ballistic expert might analyze while examining a bullet and a cartridge casing to determine the type of gun used at a crime scene. (3 marks)

Bullet: Calibre, wound ballistics (2 marks)
Cartridge casing: ammunition stamps (1 mark)

6. Police officers are trained to shoot at centre mass when faced with lethal threat encounters. Centre mass is considered to be the centre of the chest or the breastbone region. Use your knowledge of wound ballistics to explain why they are trained to aim for this part of the human body. (2 marks)

A hit to centre mass is easiest to achieve when the officer is under stress. It has the greatest chance of incapacitating a suspect who is trying to cause bodily harm to the officer or other individuals. The wound cavity would likely involve massive injury of the heart or other vital organ or cause the rib cage to shatter leading to further injury. Organs are relatively inelastic and are therefore easy to injure. This type of hit may also affect the spinal column.

A hit in the cranial area may be just as effective, but the likelihood of missing the target is great, making this tactic extremely risky when public safety is considered.

Use the following information to answer question 7.

A 20-year old male armed with a handgun shot a rival gang member twice at close range. The victim suffered two large gunshot wounds—one to his left arm and a second wound that severed the large femoral artery in the upper portion of his left leg. Each bullet expanded upon entry and neither bullet exited the victim's body.

7. a. Identify the type of bullet that likely caused these injuries. Explain your choice. (2 marks).

The bullet used was likely a hollow point because such bullets mushroom (expand) upon entry to make a large wound cavity with increased destruction likely

- b. Explain which of the two gunshot wounds most likely caused the victim's death. (2 mark)

The gunshot wound to the upper leg likely caused the victim's death. It severed a large artery, which likely caused extensive blood loss.

Name:

8. Discuss how the injuries to President Kennedy and Governor Connally would have been different if Lee Harvey Oswald had used a handgun instead of a rifle. (3 marks)

The injuries would have been less severe to the President and Governor if Oswald had used a handgun. Handgun bullets are propelled at lower velocity and, therefore, have less kinetic energy than rifle bullets.

Lesson C. Testing for Gunshot Residue

Use the information below to answer questions 9 and 10.

A piece of gauze turns a scarlet red colour after being swabbed over the hands of a murder suspect.

9. a. Before it was swabbed over the suspect's hands, what substance was the gauze dipped into? (1 mark)

A dilute hydrochloric acid

b. What specific element was detected in this test?. (1 mark)

lead

c. Identify whether the element detected in this test was from a propellant or a primer. Justify your choice. (2 marks)

Primer - lead is found only in primer

10. Why is the Scanning Electron Microscope (SEM) being used more frequently in GSR testing? (2 marks)

Any of the following for one mark each.

- The scanning electron microscope can detect and confirm any type of element, ion, or compound from gunshot residue that may be found on a suspect.
- most reliable method
- low rates of false positive results
- can analyze residues up to 12 hrs after crime

Both the Paraffin test and Harrison-Gilroy test are performed on an armed robbery suspect. The paraffin test is positive for nitrites, and the Harrison-Gilroy test is positive for antimony. X-ray analysis with a scanning electron microscope confirms the results of both tests

Both the Paraffin test and Harrison-Gilroy test are performed on an armed robbery suspect. The paraffin test is positive for nitrites, and the Harrison-Gilroy test is positive

Name:

for antimony. X-ray analysis with a scanning electron microscope confirms the results of both tests.

11. Can the combined results of these three tests ensure a conviction in court? (3 marks)

Not necessarily - positive GSR results never alone prove the guilt of a suspect. The paraffin test detects nitrites, which is in smokeless powder, and the Harrison-Gilroy test detects antimony, which is in the primer of the cartridge. These results, plus the fact that they have been confirmed by the scanning electron microscope, make it more plausible to a jury the conclusion that the suspect had recently used a firearm. It is possible that the suspect merely touched a recently used gun or been contaminated from another source. Must have other valid pieces of evidence to support this.

Lesson D. Case Study Involving Forensic Ballistics

Case Study #1: The Beltway Snipers

12. Explain why no intact bullets were found at any of the Beltway sniper shootings. (3 marks)

- No intact bullets were found in the victims because high velocity bullets explode into bullet fragments.
- High-velocity bullets explode into fragments when they hit bone and/or other tissue.

Military snipers are trained to shoot at and hit targets from great distances, typically 500 to 1000 metres.

13. Considering the range at which most of the shootings took place, why have some military experts criticized the media's use of the label *Beltway Sniper*? (3 marks)

- The Beltway sniper's victims were shot at distances of 100 metres or less.
- The name Beltway sniper was likely criticized because the shots were taken from distances far less than those for which military snipers are trained. Hitting a target at less than 500 metres is not difficult with an assault rifle

Case Study #2: Distance, Shielding, and Movement Critical Incident Response

14. A security guard arrests a man at a shopping mall after he is caught trying to steal clothing. When police arrive, they suspect he is one of the three suspects from the convenience store robbery and murder. How could GSR testing help to confirm their suspicions? (3 marks)

- Gunshot residue (GSR) testing could be completed on the suspect to determine if he has fired a gun recently.
- GSR testing would identify any of the unique particles from gunpowder or primer that are

Name:

released when a gun is fired.

15. Describe three reasons police officers confronting an armed suspect should know the type of firearm the suspect is carrying (3 marks)

- Helps them to identify the type of shielding (cover) they need.
- Helps to determine a safe distance at which the suspect can be engaged.
- Helps to determine the size of the area that needs to be cordoned off to protect the public